



NESA NATIONAL EXAMINATION
AND SCHOOL INSPECTION
AUTHORITY

Biology and Health Sciences I

001

16/07/2025

08:30 AM – 11:30 AM

**SUBJECT: BIOLOGY AND HEALTH
SCIENCES I**

DURATION: 3 HOURS

Names:

Index number:

S3 QUESTIONS and ANSWERS BOOKLET

**ORDINARY LEVEL NATIONAL EXAMINATIONS,
2024-2025**

Instructions to candidates:

- 1) Write your names and index number on the answer booklet as written on your registration form.
- 2) Do not open this question paper until you are told to do so.
- 3) This paper consists of **THREE** sections: **A**, **B** and **C**.

Section A: Attempt **all** questions. **(55 marks)**

Section B: Attempt any **three** questions.
(30 marks)

Section C: This section is **Compulsory**.
(15 marks)

- 4) Use only a **blue** or **black** pen.
- 5) For all multiple choice questions, circle a letter corresponding to the correct answer (key)

FOR EXAMINER'S USE ONLY

Questions	Marks	Questions	Marks
1		16	
2		17	
3		18	
4		19	
5		20	
6		21	
7		22	
8		23	
9		24	
10		25	
11		26	
12		27	
13		28	
14		29	
15		30	

TOTAL MARKS...../100



Section A: Attempt all questions. (55 marks)

For all multiple choice questions, circle a letter corresponding to the correct answer.

- 1) Why are flowering plants important to the environment? **(1 mark)**
- a) They provide oxygen.
 - b) They prevent erosion.
 - c) They help in the production of food and oxygen.
 - d) They absorb carbon dioxide only.
- 2) Which of the following is the primary characteristic that distinguishes the phylum Arthropoda from other animal phyla? **(1 mark)**
- a) Radial symmetry.
 - b) Exoskeleton made of chitin.
 - c) Presence of a backbone.
 - d) Bilateral symmetry.
- 3) The total magnification of a microscope is calculated by multiplying the magnification of the ocular lens by the magnification of the objective lens. If the ocular lens magnifies 10x and the objective lens magnifies 40x, what is the total magnification? **(1 mark)**
- a) 4x
 - b) 4000x
 - c) 400x
 - d) 40x
- 4) Which of the following organelles is responsible for protein synthesis in both plant and animal cells? **(1 mark)**
- a) Golgi apparatus.
 - b) Endoplasmic reticulum.
 - c) Ribosomes.
 - d) Lysosomes.

- 5) Match the nutrient in column **A** with its primary function in column **B**. **(5 marks)**

A: Nutrient	B: Function
i) Carbohydrates	a) Provide energy for the body.
ii) Proteins	b) Support immune function and cell repair.
iii) Fats	c) Aid in the formation of bones, teeth, and blood.
iv) Vitamins	d) Provide energy and help in the absorption of fat-soluble vitamins.
v) Minerals	e) Facilitate various biochemical processes in the body.

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- 6) State whether each of the following statements is **True** or **False**

a) In passive transport, molecules move from an area of low concentration to an area of high concentration. **(1 mark)**

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b) Osmosis is a type of facilitated diffusion that requires a carrier protein. **(1 mark)**

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c) Active transport moves molecules against their concentration gradient using energy, whereas passive transport moves molecules down their concentration gradient without energy. **(1 mark)**

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- 7) Circle the letter corresponding to the best answer: Which of the following statements best describes the role of enzymes in biochemical reactions? **(1 mark)**

- a) Enzymes are consumed in chemical reactions and cannot be reused.
- b) Enzymes act as catalysts, speeding up the rate of biochemical reactions without being consumed.
- c) Enzymes change the direction of chemical reactions.
- d) Enzymes increase the energy required for reactions to occur.

- 8) Match each condition in **column A** with its effect on the rate of photosynthesis in **column B**. **(5 marks)**

A: Condition	B: Effect on the rate of photosynthesis
i) Increased light intensity	a) Increases the rate of photosynthesis until the saturation point is reached
ii) Increased carbon dioxide concentration	b) Can decrease the rate if too extreme
iii) Increased temperature	c) Increases the rate of photosynthesis, up to a limit
iv) Decreased light intensity	d) Decreases the rate of photosynthesis
v) Reduced water availability	e) Reduces the rate of photosynthesis significantly

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- 9) Match each plant structure in **column A** with its primary function in the transport system in **column B**. **(5 marks)**

A: Structure	B: Function
i) Xylem	a) Transport of water and minerals from the roots to other parts of the plant.
ii) Phloem	b) Transport of organic food (mainly sugars) from leaves to other parts.
iii) Root Hair	c) Absorb water and minerals from the soil.
iv) Stomata	d) Facilitate gas exchange for photosynthesis.
v) Leaves	e) Perform photosynthesis to produce food for the plant.

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10) State whether each of the following statements is **True** or **False**.

- a) In meiosis, two rounds of division (meiosis I and meiosis II) result in the production of four genetically diverse haploid cells. **(1 mark)**

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- b) Mitosis is responsible for producing genetically identical cells, whereas meiosis produces cells with half the chromosome number of the original cell. **(1 mark)**

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- c) In meiosis, crossing over occurs during metaphase I, leading to genetic recombination. **(1 mark)**

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11) Match each characteristic of heterotrophic nutrition in **column A** with its correct description in **column B**: **(4 marks)**

A: Characteristic	B: Description
i) Ingestion	a) The process of breaking down food into simpler molecules.
ii) Digestion	b) The process of absorbing nutrients into the bloodstream.
iii) Absorption	c) The intake of food into the organism's body.
iv) Egestion	d) The removal of undigested food from the body.

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12) What is the primary function of the human circulatory system? **(1 mark)**

- a) To circulate air throughout the body.
b) To transport nutrients, oxygen, and waste products.
c) To transport hormones and enzymes.
d) To protect the body from pathogens.

13) In the absence of oxygen, what process does the body rely on to produce energy during intense exercise? **(1 mark)**

- a) Aerobic respiration.
- b) Lactic acid fermentation.
- c) Alcoholic fermentation.
- d) Glycolysis.

14) Which of the following is an example of how the skin helps maintain body temperature during excessive heat? **(1 mark)**

- a) Vasodilation and sweating.
- b) Vasoconstriction and shivering.
- c) Increased blood flow to the internal organs.
- d) Decreased sweating and increased hair growth.

15) In the human eye, which part is primarily responsible for detecting light and converting it into nerve impulses? **(1 mark)**

- a) Retina
- b) Lens
- c) Cornea
- d) Pupil

16) Match the reproductive organs in **column A** with their functions in **column B**. **(4 marks)**

A: Reproductive Organ	B: Function
i) Testis	A) Production of eggs
ii) Ovary	B) Production of sperm
iii) Uterus	C) Site of fertilization
iv) Fallopian tube	D) Site where the embryo develops

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17) Which of the following best describes a virus? **(1 mark)**

- a) A microorganism that can reproduce independently.
- b) A non-living particle that needs a host cell to replicate.
- c) A type of bacteria that causes infections.
- d) A protozoan parasite that infects humans.

18) State whether each of the following statements is **True** or **False**:

a) Using condoms correctly during sexual intercourse can reduce the risk of contracting HIV and other STIs. **(1 mark)**

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b) HIV can be transmitted through casual contact, such as shaking hands or hugging **(1 mark)**

c) Limiting the number of sexual partners can reduce the risk of contracting STIs and HIV.

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d) Taking PrEP (Pre-exposure Prophylaxis) can help reduce the risk of acquiring HIV in high-risk individuals. **(1 mark)**

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e) Sharing needles for drug use is a common way that HIV can be transmitted. **(1 mark)**

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19) A family who regularly attends community health education programs is more likely to: **(1 mark)**

- a) Neglect healthy habits in favor of traditional practices.
- b) Be more aware of health risks and preventive measures.
- c) Avoid healthcare professionals due to distrust.
- d) Ignore healthcare and physical exercise.

20) When deciding whether to have a sexual relationship, which of the following should a young adult consider to make an informed and responsible decision? **(1 mark)**

- a) The level of sexual experience each partner has.
- b) The availability of birth control methods.
- c) The long-term emotional and physical consequences of the relationship.
- d) Whether the relationship is romantic or based purely on physical attraction.

21) How does the water cycle contribute to sustaining ecosystems? **(1 mark)**

- a) It increases the salinity of water bodies.
- b) It ensures water is continuously available to plants, animals, and humans.
- c) It stops the evaporation of water.
- d) It prevents the contamination of water sources.

22) How does migration affect the population growth rate of a species in a given area? **(1 mark)**

- a) Migration out of an area typically increases the local population growth rate.
- b) Migration into an area increases the local population growth rate by introducing new individuals.
- c) Migration has no effect on population growth rates.
- d) Migration always results in a decrease in population growth.

23) Which of the following human activities is most directly linked to habitat destruction? **(1 mark)**

- a) Overfishing.
- b) Recycling.
- c) Deforestation.
- d) Sustainable agriculture.

24) In which type of reproduction do offspring inherit traits from only one parent? **(1 mark)**

- a) Asexual reproduction
- b) Sexual reproduction
- c) Both asexual and sexual reproduction
- d) Budding

25) State whether each of the following statements is **True** or **False**:

- a) Pollination is the process of fertilization in flowering plants. **(1 mark)**
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- b) The pistil is the female reproductive part of a flower. **(1 mark)**
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- c) The sperm cell in pollen fuses with the egg cell in the ovule during fertilization. **(1 mark)**
- d) Fertilization results in the formation of a zygote, which later develops into a seed. **(1 mark)**
- e) In sexual reproduction, all offspring of flowering plants are genetically identical.
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Section B: Attempt any three questions. (30 marks)

26) How does predation influence the population of the prey species, and why is this not the case for scavengers? **(10 marks)**

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27) Evaluate the role of ovulation timing in the likelihood of fertilization and the factors that contribute to this process. **(10 marks)**

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28) Osmosis is an important process in plants. Justify. (10 marks)

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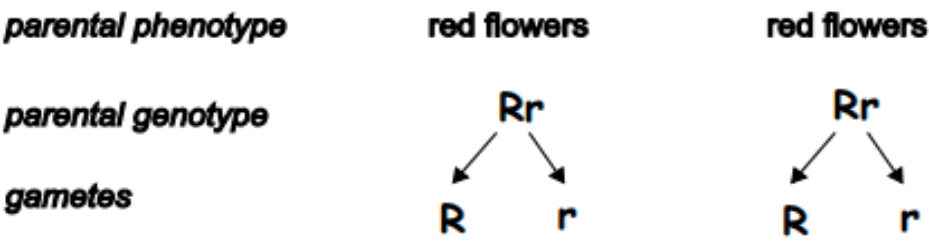
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29) A homozygous sweet pea with white flowers was crossed with a homozygous sweet pea with red flowers. All of the offspring had red flowers.

- a) State which allele of this gene is dominant. (1 mark)
- b) Two heterozygous sweet pea plants with red flowers were crossed as shown below.



Draw a Punnet square to show the genotypes that are produced in this cross. (6 marks)

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c) State the phenotypic ratio produced in this cross.

(3 marks)

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Section C: Compulsory question. (15 marks)

30) Assume you are provided with the following materials and tools: DCIP, orange, test tubes, droppers, syringes, beakers, knife.

a) Use the materials provided to prepare an extract of food sample. **(6 marks)**

b) Test the food component present in that fruit (procedure, result and conclusion) **(9 marks)**

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